

Homework 3 — NYC Taxi Analytics

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Dataset

January 2026 NYC TLC trip records

- yellow_tripdata_2026-01.parquet
- green_tripdata_2026-01.parquet

EC2 Setup

For this assignment, I used an Amazon EC2 instance as my remote compute environment. After starting the EC2 instance through the Northwestern AWS portal, I connected to it from my local machine using SSH with port forwarding. The SSH command forwarded port 8888 on my local computer to port 8888 on the EC2 instance, which allowed me to access Jupyter Notebook in my local browser.

On the EC2 instance, I ran Jupyter Notebook inside a Docker container using the Jupyter PySpark notebook image. The Docker command mapped the container's Jupyter port to the EC2 instance's port 8888 and mounted the EC2 project directory into the container. Specifically, the EC2 folder /home/ec2-user/my_project was mounted to /home/jovyan/work inside the container. This allowed files created in Jupyter to persist on the EC2 instance even if the Docker container was stopped or removed.

I used the quay.io/jupyter/pyspark-notebook:latest Docker image because it includes Jupyter and PySpark tools needed for the assignment. After starting the container, I opened Jupyter Notebook locally at <http://localhost:8888/tree> using the token printed in the Docker output.

Data Cleaning Assumptions

Cleaning rules applied to the unioned dataset:

1. Drop rows with null pickup or drop-off timestamps.
2. Drop trips with trip_distance <= 0 (zero/negative mileage is invalid).
3. Drop trips with fare_amount < 0 (negative fares treated as data errors).
4. Drop trips with total_amount < 0.
5. Drop trips where dropoff_datetime < pickup_datetime (impossible).
6. Drop trips longer than 24 hours (almost certainly meter errors or forgotten-running meters, not real rides).

Result: 3,765,161 rows before cleaning -> 3,600,946 after (4.36% removed).

Analytics Questions 1-4

Q1. Which taxi type had more trips?

Yellow had more trips: 3,562,033 vs. green 38,913 (yellow is ~92x larger).

taxi_type	trip count
yellow	3,562,033
green	38,913

Q2. What hour of day had the most pickups?

Hour 18:00 (6 PM) had the most pickups: 235,765.

pickup_hour	trips
18:00	235,765
17:00	227,332
15:00	219,306
19:00	213,794
20:00	210,273

Q3. What percentage of trips were under 2 miles?

52.12% of trips were under 2 miles (1,876,798 of 3,600,946).

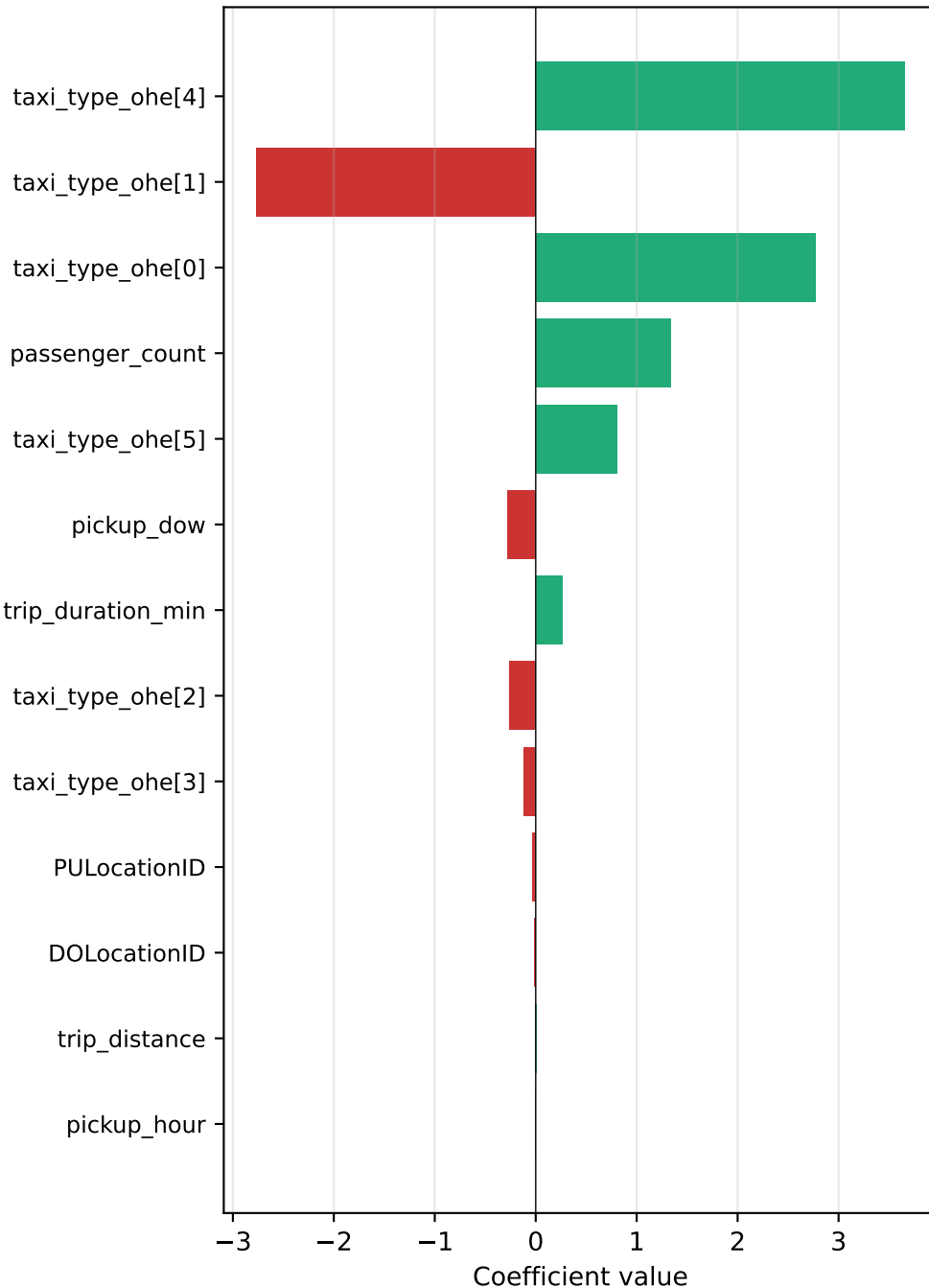
Q4. Average trip distance by taxi type

Green trips are roughly twice as long on average as yellow trips, consistent with green cabs serving outer-borough longer-haul rides.

taxi_type	avg trip_distance (mi)
green	12.989
yellow	6.705

Question 5 — Linear Regression Results

Coefficients (intercept = 19.8562)



Model performance

Training RMSE: 16.4066

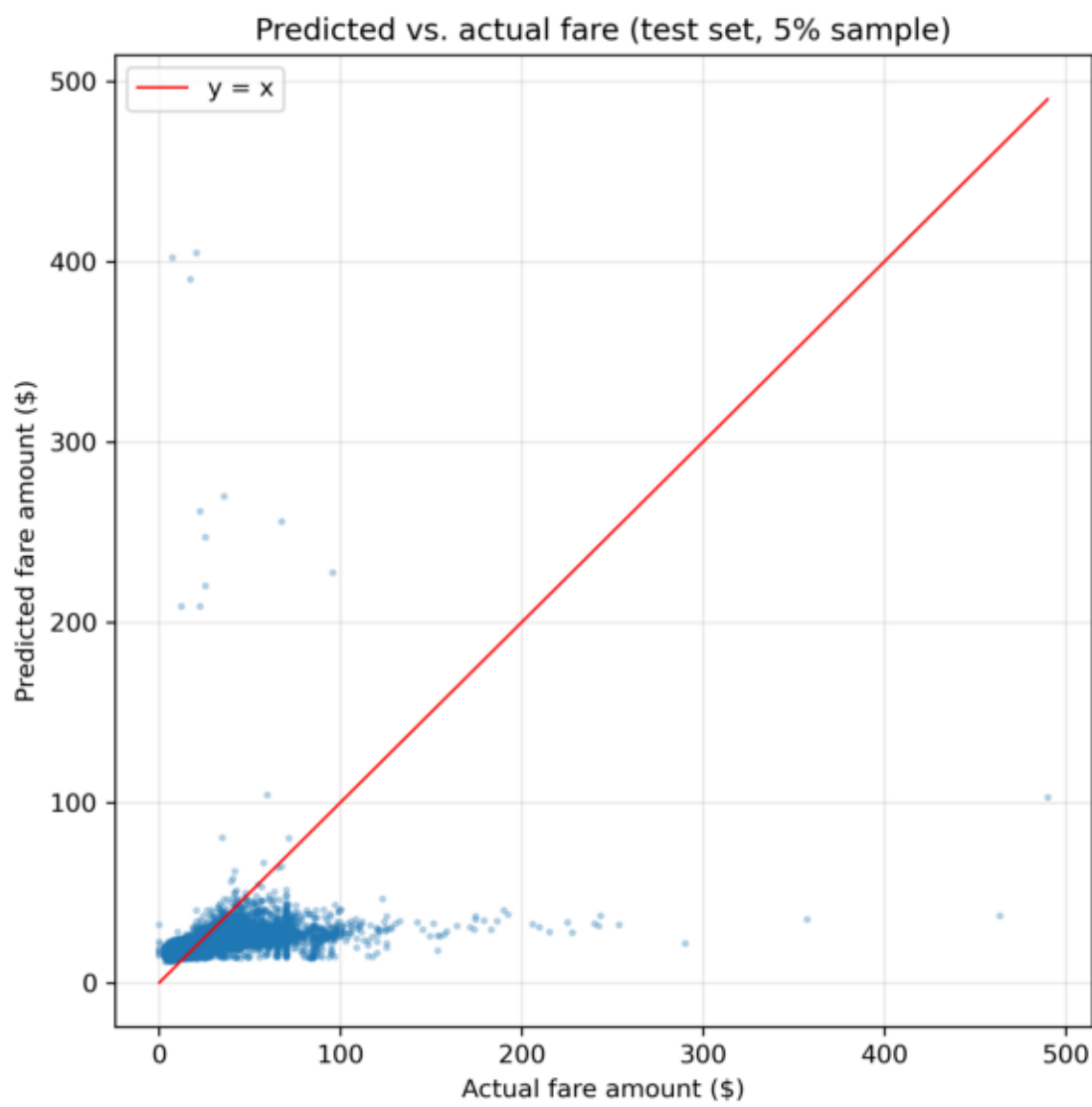
Testing RMSE: 16.4622

Train/test gap is small -> the model is not overfitting, it is simply structurally limited (see analysis below).

Predictor analysis

Predictor analysis: The dominant magnitudes are the one-hot slots for taxi_type and payment_type (the printout labels every OHE slot 'taxi_type_ohe[i]' but only two are taxi_type; the rest are payment_type). The symmetric ± 2.77 pair is a classic OHE level shift between taxi types. The most diagnostic numeric result is trip_distance = +0.0114, which is far below the real $\sim \$3/\text{mile}$ NYC rate. This signals multicollinearity with trip_duration_min (+0.2613): the two features are highly correlated and the regression splits the metered-fare signal between them. passenger_count (+1.34) is too large to be causal -- NYC meters ignore rider count -- and is really a proxy for airport/group trips. pickup_hour (~ 0) and pickup_dow (-0.28) barely move the prediction once distance/duration are controlled. PULocationID and DOLocationID are tiny because they were treated as raw integers; one-hot encoding the zones would expose the real geographic effect.

Q5 — Predicted vs. Actual Fare



Scatter of predicted vs. actual fare on a 5% sample of the test set, with a red $y = x$ reference line. Most predictions cluster between ~10–40 regardless of the actual fare, so the model systematically under-predicts long/expensive rides (actuals > ~\$100 fall well below the $y = x$ line). This is consistent with the diagnosis from the coefficients: trip_distance was deflated by multicollinearity with trip_duration_min, and the flat-rate / airport fare structure is not modelled, so the linear model cannot reach into the high-fare regime.